

Homework Submission

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Problem Statement

Solve the quadratic equation

$$x^2 - 2x + 1 = 0.$$

Key Principles and Formulas

- A quadratic equation $ax^2 + bx + c = 0$ can be solved by factoring, completing the square, or using the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- The discriminant $\Delta = b^2 - 4ac$ determines the nature of the roots:
 - $\Delta > 0$: two distinct real roots,
 - $\Delta = 0$: one real double root,
 - $\Delta < 0$: two complex conjugate roots.
- Completing the square rewrites $ax^2 + bx + c$ as $a\left(x + \frac{b}{2a}\right)^2 + \text{constant}$, making the root(s) evident.

Detailed Step-by-Step Solution

$$x^2 - 2x + 1 = 0$$

Recognize the left-hand side as a perfect square: $(x - 1)^2 = 0$

Take the square root of both sides: $x - 1 = 0$

Solve for x : $x = 1$.

Alternatively, using the quadratic formula with $a = 1$, $b = -2$, $c = 1$:

$$\Delta = b^2 - 4ac = (-2)^2 - 4(1)(1) = 4 - 4 = 0,$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a} = \frac{-(-2) \pm \sqrt{0}}{2 \cdot 1} = \frac{2 \pm 0}{2} = 1.$$

Final Answer

$$\boxed{x = 1}$$

Study Notes and Key Takeaways

- A quadratic with discriminant zero has a *double root*; the graph touches the x -axis at a single point.
- Always check whether the quadratic can be factored as a perfect square before applying the quadratic formula—it saves time and reduces arithmetic errors.
- When completing the square, ensure the coefficient of x^2 is 1; otherwise factor it out first.